

EE 446 Lab 5

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1. In the notebook, the audio waveform is transformed into time-frequency features before being passed to the keyword spotting model. What is the purpose of converting raw audio into spectrogram/MFCC-style features instead of directly using the raw time-domain waveform?

The time-domain waveform representation does not reveal as many recognizable patterns in speech signals. Converting the signal into the frequency domain highlights the harmonics, making it easier for the model to identify and group similar information. Comparing the graphs of the time domain and frequency domain shows that the loudness of a spoken word greatly affects the similarity of time-domain waveforms, while it has much less impact in the frequency domain. Since the model is intended to classify words based on their content rather than their volume, using the frequency domain improves the model's accuracy.

2. During full integer quantization, the notebook uses a representative dataset. What is the purpose of the representative dataset, and why is it important for creating an INT8 TensorFlow Lite model?

The purpose of a representative dataset is to provide a realistic range of input data so that INT8 quantization can accurately estimate the distribution of values. By using real data during calibration, the model can determine the proper value ranges before scaling FP32 values to INT8 using a scale and zero-point factor. If the estimated range is inaccurate, the quantized values may not properly represent the original data, which can reduce the accuracy of the INT8 model.

3. What does `audio_processor.get_data()` do? Refer to `input_data.py`.

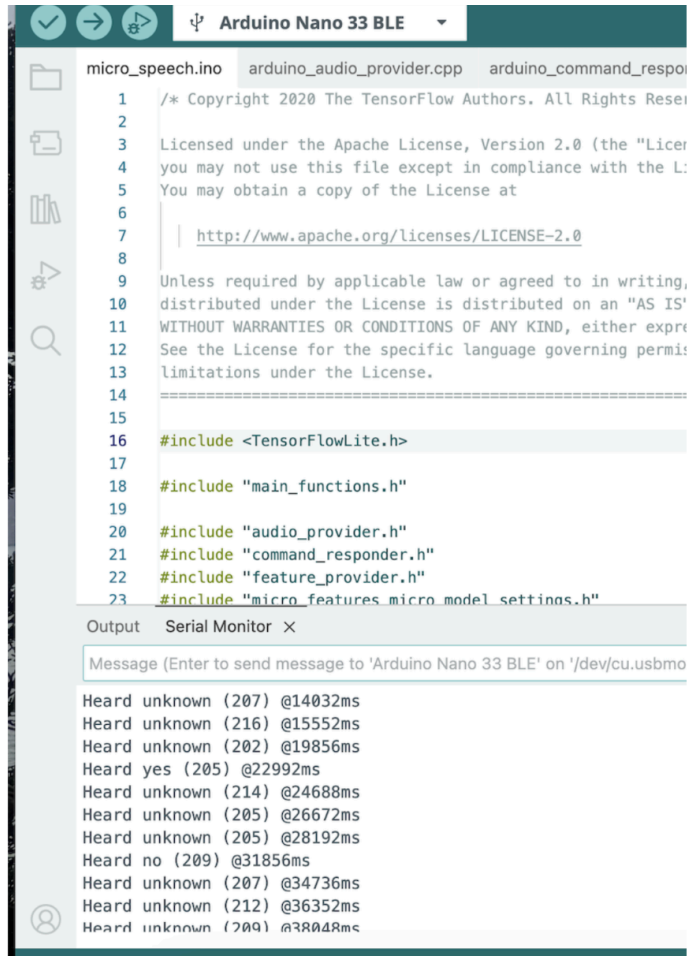
`audio_processor.get_data()` loads and prepares data for training by taking the raw audio file from the disk, and turns them into spectrograms and applies labels to the features. The results then can be put directly into the model.

4. In the final model export step, the TensorFlow Lite model is converted into a C/C++ byte array. What are `g_model` and `g_model_len`, and why do they need to be copied correctly into the Arduino source file?

`G_model` is the full binary model in hex values that make up the model, put in a byte array, and `g_model_len` is the length of the array. The interpreter needs the hex values since it reads the array as memory. If one is wrong then the model will not be loaded correctly.

5. Please include a screenshot showing the correct identification of at least one keyword, either “**YES**” or “**NO**”, as displayed in the Arduino IDE Serial

Monitor after deploying your KWS model on the Arduino Nano 33 BLE. The screenshot should be included in your submitted PDF.



The screenshot displays the Arduino IDE interface. The top toolbar shows icons for checking, running, and uploading code, along with a dropdown menu for the selected board, 'Arduino Nano 33 BLE'. The file explorer on the left lists three files: 'micro_speech.ino', 'arduino_audio_provider.cpp', and 'arduino_command_responder.h'. The main editor window shows the source code for 'micro_speech.ino', which includes a copyright notice, license information (Apache License 2.0), and several include statements for TensorFlow Lite and custom headers. The Serial Monitor at the bottom shows a series of messages indicating the model's output for various audio inputs.

```
1  /* Copyright 2020 The TensorFlow Authors. All Rights Reserved.
2
3  Licensed under the Apache License, Version 2.0 (the "License");
4  you may not use this file except in compliance with the License.
5  You may obtain a copy of the License at
6
7  http://www.apache.org/licenses/LICENSE-2.0
8
9  Unless required by applicable law or agreed to in writing, software
10 distributed under the License is distributed on an "AS IS" BASIS,
11 WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
12 See the License for the specific language governing permissions and
13 limitations under the License.
14
15
16 #include <TensorFlowLite.h>
17
18 #include "main_functions.h"
19
20 #include "audio_provider.h"
21 #include "command_responder.h"
22 #include "feature_provider.h"
23 #include "micro_features_micro_model_settings.h"
```

Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano 33 BLE' on '/dev/cu.usbmodem14101')

```
Heard unknown (207) @14032ms
Heard unknown (216) @15552ms
Heard unknown (202) @19856ms
Heard yes (205) @22992ms
Heard unknown (214) @24688ms
Heard unknown (205) @26672ms
Heard unknown (205) @28192ms
Heard no (209) @31856ms
Heard unknown (207) @34736ms
Heard unknown (212) @36352ms
Heard unknown (209) @38048ms
```